LUNAR VISION PLATFORM · INFRASTRUCTURE & DATABASE RUNBOOK

New Project Setup & Full SQL Migration Guide

Complete deployment architecture, dependency-ordered PostgreSQL migrations (01 → 17), multi-tenant RBAC & clearance isolation, Face++ 1:N biometrics, bulk data migration workflows, and verification steps.

Migrations: 17 Sequential Steps | Engine: Supabase PostgreSQL + RLS | Biometrics: Face++ V3 (US / CN) | Frontend: Vite + React 19 (Vercel)

1. Platform Architecture & Isolation Model

Lunar Vision serves two strictly isolated operational portals from one codebase, backed by Supabase PostgreSQL Row-Level Security (RLS), private Storage buckets, and server-side API endpoints:

- **Business Workspace (`/business`):** Self-registered commercial organizations with customer footfall, occupancy analytics, camera intelligence settings, incident remediation (`lv_corrective_actions`), and commercial staff access. ANPR plate watchlists, vehicle registries, Face++ search, criminal dossiers, and security dashboards are never exposed.
- **Security Agency Console (`/agency`) & Platform Admin (`/agency/admin`):** Platform Super Admins (`lv_platform_admins`) provision agencies, issue initial administrator accounts, and toggle organization feature entitlements (`lv_workspace_features`). Agency accounts include People/Vehicle/Advanced camera analytics, ANPR hotlists, Pre-Populated Vehicle Registration DB, Multi-Angle Biometric Subject Enrollment, Bulk CSV/Excel + Photo Migration, 1:1 / 1:6 / 1:N Face++ Search, and immutable audit history.
- **Six-Tier Roles & Four Clearance Tiers:** Roles span `owner` , `admin` , `supervisor` , `investigator` , `member` , and read-only `viewer` , paired with Clearance Levels `0` (General) through `3` (Confidential) and per-member `allowed_modules` (`monitor` , `analytics` , `records` , `evidence` , `vehicles` , `access`).

> **Security Scope Notice:** Non-consensual public website scraping (`crawler` and `crawled_faces`) remains permanently disabled in the agency console. All Face++ biometric indexing operates strictly on agency-uploaded authorized dossiers and consented records inside isolated per-agency FaceSets (`outer_id = lv_<workspace_uuid_hex>`).

2. Choosing Your Migration Path

Scenario	What to Run	Notes
A. Brand-New Empty Supabase Project	Run `migration-setup/FULL_MIGRATION_NEW_PROJECT.sql` once in Supabase SQL Editor (or run `ordered-migrations/01_*.sql` through `17_*.sql` in order).	Step 03 safely checks `to_regclass(...)` before revoking legacy EagleEye v1 tables, so an empty database completes cleanly with zero errors.
B. Existing Database Upgrade (e.g. error `42701: clearance already exists`)	Run diagnostic SQL below first, then execute **only missing steps** from `ordered-migrations/` in ascending numeric order.	Do **not** re-run earlier migrations that have already committed; migrations 01–13 alter existing tables and will fail if re-applied out of state.

Diagnostic Check for Existing Databases

Paste this read-only query in Supabase SQL Editor to see which milestone steps are already committed:

```
select
  to_regclass('public.lv_workspaces')                                  as step_01,
  to_regclass('public.lv_intel_events')                                as step_12,
  to_regprocedure('public.lv_has_scope(uuid,text,integer)')            as step_13,
  to_regclass('public.lv_agency_subjects')                             as step_14,
  to_regprocedure('public.lv_manage_workspace_access(uuid,text,text)') as step_15,
  to_regprocedure('public.lv_configure_camera_connection(uuid,bigint,text,text,text,text,integer)') as
step_16,
  to_regprocedure('public.lv_staff_can_write(uuid,text,integer)')      as step_17;
```

> **Example:** If `step_13` is populated (non-null) and `step_14` is `NULL`, skip files `01` – `13` and run **`14`, `15`, `16`, and `17`** in strict order.

3. Complete Ordered Migration Reference (Steps 01 → 17)

Every file in `ordered-migrations/` corresponds 1:1 with `supabase/migrations/` in exact execution sequence:

Step	Migration File Name	Tables, Constraints, Policies & RPC Functions Added
01	`202611100001_workspaces.sql`	Creates `lv_workspaces`, `lv_memberships`, `lv_platform_admins`, `lv_records`; enables RLS; creates `lv_create_business`, `lv_create_agency`, `lv_agency_access`.
02	`202611100002_cameras.sql`	Creates `lv_cameras` with HTTPS playback URL regex validation and workspace member read/owner write RLS.
03	`202611100003_legacy_quarantine.sql`	Revokes public/anon/auth access from legacy EagleEye v1 tables (`audit_logs`, `suspects`, `watchlist`, etc.) & blocks legacy `mugshots` bucket. *(Guarded by `to_regclass` in setup bundle.)*
04	`202611100004_record_status.sql`	Adds `status` workflow (`open`, `in_review`, `resolved`) on `lv_records`, `lv_record_activity` audit table, and optimistic `lv_change_record_status`.
05	`202611100005_vehicle_watchlist.sql`	Creates `lv_vehicle_watchlist` with normalized plate constraint, active-plate unique index, and `lv_retire_vehicle_watch`.
06	`202611100006_workspace_governance.sql`	Adds `active`, `entitlement`, `organization_details` to `lv_workspaces`; adds `lv_workspace_activity`, `lv_workspace_active`, `lv_configure_agency`, `lv_workspace_staff`.
07	`202611100007_video_observations.sql`	Creates `lv_video_observations` recording frame timestamps, camera/file source, and detection summaries per active workspace.
08	`202611100008_corrective_actions.sql`	Creates `lv_corrective_actions` for business incident remediation deadlines & `lv_complete_corrective_action` RPC.

Step	Migration File Name	Tables, Constraints, Policies & RPC Functions Added
09	`202611110001_command_map.sql`	Adds validated `latitude` / `longitude` columns & update policies on `lv_cameras` and investigation records.
10	`202611110002_agency_operations.sql`	Creates `lv_agency_rules`, `lv_agency_alerts`, `lv_agency_evidence`, `lv-agency-evidence` bucket (25 MB), rule-trigger evaluator `lv_observation_rules`, and `lv_transition_alert`.
11	`202611110003_operation_reliability.sql`	Adds optimistic concurrency `revision` columns, idempotent observation `request_id` via `lv_save_observation`, and client SHA-256 `lv_register_evidence`.
12	`202611120001_video_intelligence.sql`	Expands roles (`supervisor`, `investigator`); adds `lv_workspace_features`, `lv_camera_intel_settings`, `lv_intel_events`, `lv_vehicle_registry`, `lv_registry_imports`; expands watchlist categories/photos; adds real-time ANPR matcher `lv_record_intel_events` & `lv-intel-snapshots` / `lv-vehicle-photos` buckets.
13	`202611120002_workspace_access.sql`	Adds `viewer` role, `clearance` (0–3), `allowed_modules`, `stream_type` (`video` / `hls` / `mjpeg`), `lv_has_scope`, `lv_set_staff_access`, restrictive clearance policies & write-guard triggers.
14	`202611120003_enterprise_enrollment_migration.sql`	Adds `biometric_enrollment` & `face_search` toggles; creates `lv_subject_migrations`, `lv_agency_subjects`, `lv_subject_photos`, `lv_face_search_history`, `lv-subject-photos` bucket (4 MB), clearance helper `lv_subject_clearance_ok`, migration/search RPCs, and bulk `lv_import_watchlist`. *(Idempotent.)*
15	`202611120004_preserve_staff_access.sql`	Updates `lv_manage_workspace_access` so legacy role changes retain existing clearance levels and module arrays.
16	`202611120005_camera_connections.sql`	Adds `lv_configure_camera_connection` RPC for safe revision-checked camera endpoint, format, and clearance updates.
17	`202611120006_viewer_write_guards.sql`	Adds `lv_staff_can_write` and trigger/storage guards blocking direct inserts, alerts, and uploads from read-only `viewer` accounts.

4. Private Storage Buckets & Retention Policies

All operational buckets are private (`public = false`) and accessed via short-lived signed URLs after membership + module + clearance validation:

Bucket ID	Max Size	Allowed MIME Types	Tenant Folder Prefix & Scope
`lv-agency-evidence`	25 MB	JPEG, PNG, PDF, MP4	`<workspace_uuid>/<uploader_uid>/...` — immutable once registered via `lv_register_evidence`.

Bucket ID	Max Size	Allowed MIME Types	Tenant Folder Prefix & Scope
`lv-intel-snapshots`	2 MB	JPEG, PNG, WebP	`<workspace_uuid>/...` — restricted to callers cleared for the parent `lv_intel_events` camera.
`lv-vehicle-photos`	2 MB	JPEG, PNG, WebP	`<workspace_uuid>/...` — requires agency membership + `vehicles` scope (`watchlist_write` to upload).
`lv-subject-photos`	4 MB	JPEG, PNG, WebP	`<workspace_uuid>/subjects/...` & `/probes/...` — requires agency membership + biometric/search permission.

5. Supabase Auth, Environment Variables & API Endpoints

5.1 Supabase Authentication Redirect Configuration

In Supabase Dashboard → **Authentication** → **URL Configuration**, set the Site URL and whitelist these exact Redirect URLs:

- **Local Vite:** `http://localhost:3000/business`, `http://localhost:3000/agency`, `http://localhost:3000/agency/admin`
- **Vercel Production:** `https://<your-domain>/business`, `https://<your-domain>/agency`, `https://<your-domain>/agency/admin`

If using Google OAuth without SMTP email delivery, configure a Google Cloud OAuth client inside Supabase Auth Providers, keep email confirmation enabled, and set `VITE_GOOGLE_SIGN_IN_ENABLED=true`. Set `VITE_EMAIL_AUTH_FLOWS_ENABLED=false` if self-serve email sign-up/recovery should be hidden.

5.2 Environment Variables (`.env.local` locally / Vercel Settings in production)

Variable	Scope	Configuration Details
`VITE_WORKSPACE_PORTAL_ENABLED`	**Client**	Must be `true` to enable Supabase portals (especially in `vercel` build mode per `services/supabase.ts`).
`VITE_SUPABASE_URL`	**Client**	Project API URL from Supabase Settings → API.
`VITE_SUPABASE_ANON_KEY`	**Client**	Public anonymous key (safe for browser; all security is enforced by RLS).
`VITE_MAPBOX_ACCESS_TOKEN`	**Client**	Mapbox public access token for Command Center and Global Map tiles.
`VITE_FACE_COMPARISON_ENABLED`	**Client**	Set `true` to activate 1:1 Face Compare, 1-vs-6 Gallery, Subject Enrollment & 1:N Face++ Search UI.
`VITE_GOOGLE_SIGN_IN_ENABLED` / `VITE_EMAIL_AUTH_FLOWS_ENABLED`	**Client**	Authentication UI toggles (`true` / `false`).
`FACEPP_API_KEY`	**Server-only**	Face++ API Key. Keep unprefixed (never `VITE_`) so secrets never bundle into browser JS.
`FACEPP_API_SECRET`	**Server-only**	Face++ API Secret paired with `FACEPP_API_KEY`.
`FACEPP_REGION`	**Server-only**	`US` (`api-us.faceplusplus.com/facepp/v3`, default) or `CN` (`api-cn.faceplusplus.com/facepp/v3`).
`STAFF_SUPABASE_SERVICE_ROLE_KEY`	**Server-only**	Supabase `service_role` key for `/api/agency-accounts` (falls back to `SUPABASE_SERVICE_ROLE_KEY`).
`AGENCY_SETUP_ORIGIN`	**Server-only**	Trusted exact HTTPS origin (`https://<your-domain>` on Vercel, or `http://127.0.0.1:3000` locally).

5.3 Serverless / Dev API Routes

Both local Vite middleware (`vite.config.ts`) and Vercel serverless routes (`api/*.ts`) expose three authenticated JSON/multipart endpoints (each validates the caller JWT and agency membership before acting):

- `POST /api/agency-accounts` (`api/agency-accounts.ts`): Issues staff accounts with confirmed email + one-time recovery link without sending automatic password-reset emails to existing users.
- `POST /api/face-compare` (`api/face-compare.ts`): 1:1 and 1-vs-6 ad-hoc Face++ comparison with concurrency retry backoff.

- `POST /api/face-intelligence` (`api/face-intelligence.ts`): Isolated per-agency FaceSet management (`enroll_face`, `search_database` 1:N lookup + clearance hydration + `lv_log_face_search` audit log, and `remove_face`).

6. Initial Bootstrapping, Cameras & Bulk Migration Workflows

1. **Create & Confirm Super Admin User:** Sign up or create the primary platform administrator account in Supabase Dashboard → Authentication → Users (ensure `email_confirmed_at` is confirmed).
2. **Promote Super Admin in SQL Editor:**

```
insert into public.lv_platform_admins (user_id)
select id from auth.users where lower(email) = 'superadmin@yourdomain.gov'
on conflict do nothing;
```

3. **Provision Organizations:** Sign in at `/agency/admin` to create Security Agency workspaces, assign an initial Owner/Admin, and toggle per-organization features (including `anpr`, `vehicle_registry`, `security_intelligence`, `biometric_enrollment`, `face_search`).
4. **Connect Live Cameras:** Browser playback requires HTTPS MP4/WebM, HTTPS HLS (H.264/AAC), or HTTPS MJPEG streams with CORS headers matching your app origin. RTSP/ONVIF LAN feeds must be bridged through a local/edge media gateway before entering the HTTPS URL in *Live cameras*.
5. **Bulk Criminal Dossier & Mugshot Migration:** In *Enroll subject → Bulk Data & Photo Migration*, upload a CSV/XLSX dataset alongside a multi-select folder of mugshots/pictures. Records are matched by `photo_filename` or `subject_code`, compressed client-side when over 1.2 MB, indexed via `/api/face-intelligence` into the agency's isolated Face++ FaceSet, stored in `lv-subject-photos`, and tracked in `lv_subject_migrations` with downloadable error CSV reports.
6. **Bulk Watchlist & Vehicle Registry Imports:** Use *Vehicle watchlist → + Bulk Migrate Plate Watchlist* (CSV/XLSX + plate-named vehicle photos via `lv_import_watchlist`) and *Vehicle registry → Bulk CSV/Excel Import* (1,000-row streaming chunks via `lv_import_registry`).

7. Post-Migration Verification & Troubleshooting

7.1 Verification Checklist (Run Locally Before Deploying)

```
npm ci
npx tsc --noEmit
npm run test:agency
node tests/workspace-access.integration.mjs
npx vitest run api/face-compare.test.ts api/face-intelligence.test.ts api/staff-accounts.test.ts
services/videoIntelligence.test.ts components/FaceComparison.test.tsx
components/VideoIntelligenceWorkflows.test.tsx pages/WorkspacePortal.test.tsx --testTimeout=15000
npm run build -- --mode vercel
```

7.2 Live Supabase SQL Smoke Check

```
-- Verify all 4 private buckets and latest feature constraints exist cleanly:
select id, public, file_size_limit from storage.buckets
where id in ('lv-agency-evidence','lv-intel-snapshots','lv-vehicle-photos','lv-subject-photos')
order by id;
```

7.3 Troubleshooting Common Errors

Symptom / Error Code	Root Cause & Exact Fix
`42701: column "clearance" of relation "lv_memberships" already exists`	`202611120002_workspace_access.sql` is already committed. Skip Steps `01` – `13`, run the Section 2 diagnostic SQL, and execute only missing steps (`14` – `17`).
`42P01: relation "public.audit_logs" does not exist`	You ran raw `202611100003_legacy_quarantine.sql` on a brand-new empty DB. Use `migration-setup/FULL_MIGRATION_NEW_PROJECT.sql` or `ordered-migrations/03_...sql` instead (both safely guard via `to_regclass`).
Face++ `AUTHENTICATION_ERROR` or `INVALID_OUTER_ID`	Verify `FACEPP_API_KEY` , `FACEPP_API_SECRET` , and matching `FACEPP_REGION` (`US` vs `CN`) in server env vars (restart `npm run dev` locally after updating `.env.local`).
Portal shows setup disabled or static fallback	Ensure `VITE_WORKSPACE_PORTAL_ENABLED=true` , `VITE_SUPABASE_URL` , and `VITE_SUPABASE_ANON_KEY` are set in Vercel Environment Variables and trigger a fresh redeploy.
Camera loads in VLC but fails browser AI/ANPR	Browsers cannot play raw RTSP or extract canvas pixels across origins without CORS. Proxy the stream through an HTTPS HLS/MJPEG gateway with `Access-Control-Allow-Origin` set for your domain.